

OCEAN-SHIELD — MARITIME ENVIRONMENTAL ANALYTICS

SIH26143 PROTOTYPE • ANALYST-REVIEW CASE SUMMARY

AUTOMATED SCREENING OUTPUT — NOT A STATUTORY NOTICE OR FINDING OF LIABILITY
Requires calibrated imagery, source provenance, qualified analyst review, and competent-authority action.

Dossier Ref:	ICG-MEP-DOSSIER-20260906-GULF	Date of Issue:	06 Sep 2026, 11:29 UTC
Maritime Sector:	Gujarat Offshore / Northern Arabian Sea	Review status:	SCREENING LEAD — REVIEW REQUIRED
Target Vessel:	MT NEPTUNE GLORY	IMO / MMSI:	9384722 / 636019482
Flag State:	Liberia	Attribution rank:	92.2% (MODEL SCORE)

1. SATELLITE RADAR REMOTE SENSING (SAR) SCREENING

Sensor Platform:	Sentinel-1A C-Band SAR	Acquisition Timestamp:	2026-09-04 13:10:00 UTC
Swath Mode & Pol:	Interferometric Wide (IW) Swath / VV (Co-polarization optimal for sea surface roughness)	Ground Resolution:	10.0 meters/pixel
Observed Slick Centroid:	22.4700° N, 69.2323° E	Total Spill Surface Area:	1.07 km²
Estimated Discharge Mass:	23.5 Metric Tonnes	Lookalike Rejection Score:	75.0% Mineral Crude Confidence
ADIOS Weathering State:	Emulsified Petroleum Hydrocarbon	Evaporative Loss / Mousse:	28.4% Evaporated / 62.1% Water Uptake

2. REVERSE LAGRANGIAN HYDRODYNAMIC DRIFT ANALYSIS

Using an advection-diffusion Lagrangian numerical formulation integrated with real-time oceanographic current vector fields ($\Delta u=0.28$ m/s, $\Delta v=0.18$ m/s) and Ekman surface windage (3.2% Stokes drift factor with 15° Coriolis deflection), the slick plume trajectory was hindcasted backwards in time. The model estimates the following **candidate release origin**; positional and temporal uncertainty must be quantified against authoritative current, wind, and imagery inputs before operational use:

Spill Origin Point (x_0, y_0):	22.38500° N, 69.11500° E	Calculated Slick Age:	10.5 Hours prior to SAR pass
Estimated Discharge Time:	T – 10.5h (06 Sep 2026 approx 02:40 IST)	Total Hydrodynamic Drift:	14.80 km displacement

3. AIS CORRELATION & KINEMATIC BEHAVIORAL ANOMALIES

Analysis of **14 vessels** operating in the sector filtered down to **5 corridor candidates**. Vessel **MT NEPTUNE GLORY** was ranked as the highest model-scored lead based on spatio-temporal proximity and kinematic features:

Evaluation Metric	Observed Telemetry Value	Anomaly Assessment	Score Weight
Closest Point of Approach (CPA)	0.08 NM (0.15 km)	Modelled proximity to candidate origin	99.9/100
Temporal Coincidence	$\Delta t = 0.00$ hours from origin	Modelled temporal alignment	100.0/100
Speed Over Ground (SOG)	Drop: 14.7 → 5.1 kts	Severe speed drop (14.7 -> 5.1 kts) matching illegal bilge purge profile	100.0/100
Vessel Risk Category	Crude / Product Tanker (DWT: 158,000 T)	High capacity oily water / slop separator	65.0/100

5. COASTAL HAZARD & BEACHING INTERCEPTION ALERT

MODELLED SHORELINE ALERT: Hydrodynamic forecasting estimates possible slick impact along the coastline within 11.2 hours. Interception coordinates: 22.5800° N, 69.4500° E. This scenario output should be reviewed by the responsible incident commander before any response decision.

5. ANALYST NEXT STEPS

- 1. Obtain original calibrated SAR/EO products and preserve their source metadata and hashes.
- 2. Re-run drift with authoritative, time-aligned current and wind fields and report uncertainty bounds.
- 3. Verify AIS completeness, vessel identity, and timing against the original provider export; obtain independent corroboration before contacting vessel **MT NEPTUNE GLORY (IMO: 9384722)**.
- 4. Refer any enforcement decision to the competent authority and applicable law.

CERTIFICATE OF AUTHENTICITY UNDER SECTION 65B OF THE INDIAN EVIDENCE ACT, 1872 / SECTION 63 OF BHARATIYA SAKSHYA ADHINIYAM (BSA), 2023:
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